Explainable AI For Early Fetal Distress Detection: A SHAP-Enhanced CTG Analysis

Minna Fathima
School of Computer Science and Engineering
Vellore Institute of Technology
Chennai, India
minna.fathima2021@vitstudent
.ac.in

***Abstract*—— Accurate assessment of fetal health is essential for reducing preventable risks associated with pregnancy and childbirth. This study addresses the problem of automated fetal health classification using Cardiotocography (CTG) data obtained from Kaggle. A systematic machine learning approach is employed, involving data preprocessing, feature scaling, and handling class imbalance using SMOTE, followed by classification using a Random Forest model. To enhance transparency and reliability in clinical decision-making, SHAP-based explainable AI techniques are integrated to interpret model predictions. The proposed approach achieves an overall accuracy of approximately 91%, with strong ROC-AUC scores across all classes, demonstrating effective discrimination between normal, suspect, and pathological conditions. The results indicate high reliability in identifying critical pathological cases, which is crucial for early intervention. The study concludes that combining robust classification techniques with interpretability methods can provide a dependable and transparent framework for fetal health assessment. These findings highlight the potential of such systems to support healthcare professionals in making informed decisions and improving prenatal care outcomes, particularly in resource-constrained environments.**

I. Introduction

Fetal health assessment is a fundamental component of prenatal care, as it plays a vital role in identifying potential risks that may affect the well-being of both the mother and the fetus. Complications arising during pregnancy, if not detected at an early stage, can lead to severe outcomes including fetal distress, preterm birth, or even mortality. Therefore, continuous monitoring and timely diagnosis are essential to ensure safe pregnancy outcomes. One of the most widely used techniques for monitoring fetal condition is Cardiotocography (CTG), which records fetal heart rate (FHR) and uterine contractions over a period of time. These physiological signals provide critical information about the oxygen supply and neurological status of the fetus.

Despite its widespread use, the interpretation of CTG data is often complex and requires significant clinical expertise. The patterns observed in CTG signals can vary widely depending on multiple factors, making manual analysis both time-consuming and subjective. Different clinicians may interpret the same CTG trace differently, leading to inconsistencies in diagnosis. This subjectivity introduces a need for automated and standardized approaches that can assist healthcare professionals in making accurate and consistent decisions.

II. Literature Survey

[1] Kaya et al. (2025) proposed a hybrid deep learning framework that utilizes AlexNet for feature extraction combined with metaheuristic optimization algorithms such as Whale Optimization Algorithm and War Strategy Optimization for feature selection. The extracted features are then classified using an SVM model to improve fetal health prediction accuracy. By transforming CTG signals into 2D representations, the study enhances the model's ability to capture complex patterns. However, the approach involves high computational complexity and requires further validation on larger and diverse datasets to ensure generalization in real-world clinical scenarios.

[2] Jain et al. (2025) introduced a Vision Transformer (ViT)-based deep learning model for analyzing ultrasound images in fetal diagnostics. The transformer architecture enables better feature representation and improves classification performance compared to traditional convolutional neural networks. The study demonstrates the potential of transformer-based models in predicting fetal health conditions and severity levels. This work emphasizes the importance of validating predictive models across different populations. However, the approach requires a large amount of labeled imaging data and significant computational resources, and its real-time clinical applicability remains limited.

[3] Rezende et al. conducted an external validation of the Fetal Medicine Foundation (FMF) competing risks model for predicting preeclampsia using maternal clinical and biomarker data. The study evaluates the model's performance in a Brazilian population and highlights its effectiveness outside the original development setting. This work emphasizes the importance of validating predictive models across different populations. However, the findings may not generalize globally, as prediction accuracy depends on data availability and regional differences in clinical characteristics.

[4] Yenkikar et al. proposed a multi-modal AI framework that integrates Siamese Neural Networks with Few-Shot Learning to analyze multiple fetal health data sources. This approach improves early risk assessment by enabling the model to learn from limited labeled data, making it suitable for medical datasets with scarcity of annotations. While the framework enhances adaptability and prediction accuracy, it introduces architectural complexity and requires extensive validation on large-scale clinical datasets.

[5] Ettiyan and Geetha developed a Heap Optimization-based Multi-LSTM model integrated with wearable device data for continuous maternal and fetal health monitoring. The system leverages real-time sensor data to improve monitoring accuracy and provide continuous tracking of physiological parameters. This approach supports proactive healthcare management; however, it heavily depends on sensor reliability and data quality. Additionally, large-scale real-world validation is required for practical clinical deployment.

[6] Abunadi applied machine learning techniques with feature engineering to analyze cardiotocography (CTG) signals for early detection of fetal anomalies. The study demonstrates that carefully engineered features significantly improve classification performance and aid in identifying abnormal fetal conditions. While the approach is effective, it relies on conventional machine learning models, which may limit the ability to capture highly complex patterns in CTG data. Further validation on diverse datasets is also necessary.

[7] Kapila and Saleti proposed an ensemble machine learning model that combines both feature selection and feature extraction techniques for CTG data analysis. By integrating multiple models, the system improves classification accuracy and robustness in fetal health prediction. However, the ensemble approach increases computational complexity and may depend heavily on dataset characteristics, requiring further evaluation across varied clinical datasets.

[8] Mushtaq and Veningston developed an interpretable deep learning model for fetal health classification using CTG data, incorporating techniques such as batch normalization and dropout to enhance generalization. The model achieved high accuracy and sensitivity, outperforming several traditional machine learning models. Additionally, it provides interpretability, which is crucial for clinical decision-making. Despite its effectiveness, the model requires large datasets and high computational power, and further real-world validation is needed.

[9] Subasi et al. applied a Bagging Ensemble Classifier to classify CTG data for predicting fetal health risks. By combining multiple decision tree models, the study improves prediction accuracy and model stability compared to single classifiers. The ensemble approach enhances robustness; however, it reduces interpretability and increases computational cost. Further validation on larger and more diverse datasets is required.

[10] Zhao et al. proposed a CNN–RNN hybrid deep learning framework for classifying intrapartum CTG signals. The CNN component extracts spatial features, while the RNN captures temporal dependencies in fetal heart rate signals This combined approach improves the model's ability to analyze both spatial and temporal aspects of CTG data, leading to better classification performance. However, the model requires large labeled datasets and significant computational resources, and further validation is needed for clinical applicability.

III. PROPOSED METHOD0LOGY

The proposed system presents an intelligent and interpretable framework for fetal health assessment using cardiotocography (CTG) data. The methodology integrates machine learning with explainable artificial intelligence to provide accurate predictions along with clinical interpretability. The system is designed as a layered architecture that includes data acquisition, preprocessing, model training, prediction, explainability, and reporting. By combining automated analysis with human-understandable explanations, the system aims to support clinicians in making informed and timely decisions.

The methodology begins with the utilization of the CTG dataset obtained from Kaggle, which contains multiple fetal health parameters such as baseline fetal heart rate, variability measures, and deceleration patterns. The dataset is first subjected to preprocessing, where irrelevant or redundant features are removed, and the target variable is encoded into three classes: Normal, Suspect, and Pathological. To ensure consistency and improve model performance, feature scaling is applied using StandardScaler, which standardizes the input data. Additionally, to address class imbalance commonly present in medical datasets, the SMOTE technique is employed during the training phase to generate synthetic samples for minority classes, thereby enhancing the model's ability to learn from all categories effectively.

Following preprocessing, a Random Forest classifier is trained on the balanced dataset. Random Forest is selected due to its robustness, ability to handle non-linear relationships, and effectiveness in multi-class classification problems. The model learns complex interactions among CTG features and outputs both class predictions and probability scores for each category. The performance of the model is evaluated using standard metrics such as accuracy, ROC-AUC, and confusion matrix to ensure reliability and generalization.

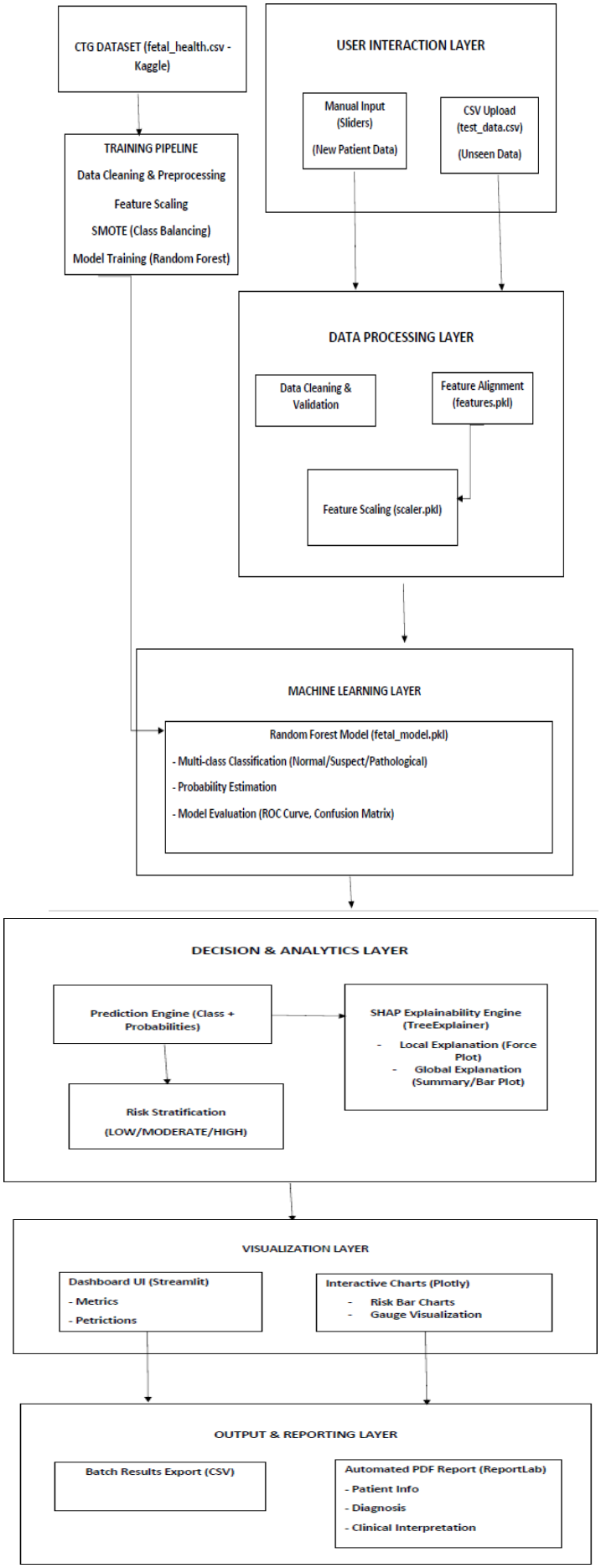

Fig 1. Architecture Diagram

Once trained, the model and associated preprocessing components, including the scaler and feature list, are serialized and stored for deployment.

In the deployment phase, the system accepts input through a user-friendly interface built using Streamlit. Users can either provide real-time patient data using sliders or upload batch data in CSV format for multi-patient analysis. The input data undergoes the same preprocessing steps as the training data, including validation, feature alignment, and scaling, to maintain consistency. The processed data is then fed into the trained model to generate predictions and corresponding probability scores, which indicate the likelihood of each fetal health condition.

Accuracy, precision, recall, specificity, F1-score, Matthews Correlation Coefficient (MCC), and the area under the ROC and Precision-Recall curves are among the metrics used to evaluate the model on a held-out test dataset. Sensitivity is specifically given priority in order to reduce the number of false negatives, which is crucial when it comes to cancer screening. Plotting calibration curves is another way to evaluate how reliable the predicted probabilities are.

To enhance transparency and trust in the model's predictions, an explainability layer based on SHAP (SHapley Additive exPlanations) is integrated into the system. SHAP values provide both local and global explanations of model behavior. Local explanations illustrate how individual features influence a specific prediction, while global explanations highlight overall feature importance across the dataset. This enables clinicians to understand the reasoning behind the model's decisions, ensuring that the system functions as a reliable decision support tool rather than a black-box model.

Finally, the system includes a visualization and reporting module that presents results in an intuitive and clinically meaningful format. Interactive dashboards display predictions, risk levels, and graphical representations such as bar charts and gauge indicators. Additionally, the system generates automated PDF reports that include patient details, CTG parameters, diagnosis, probability distribution, and clinical interpretation. These reports are designed to be hospital-ready, supporting documentation and aiding in clinical communication. Overall, the proposed methodology ensures accuracy, interpretability, and usability, making it suitable for real-world healthcare applications.

IV. RESULT AND DISCUSSION

A. Results

The proposed fetal health classification system was evaluated using standard performance metrics including accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, and feature importance analysis. The model achieved an overall accuracy of **91%**, indicating strong predictive capability across all three classes: Normal, Suspect, and Pathological.

• From the classification report, Class 0 (Normal) demonstrated excellent performance with a precision of **0.97**, recall of **0.92**, and F1-score of **0.95**, confirming the model's ability to correctly identify healthy fetal conditions with minimal misclassification. Class 2 (Pathological) also exhibited strong performance, achieving a precision of **0.89**, recall of **0.91**, and F1-score of **0.90**, which is particularly important for detecting critical cases. Class 1 (Suspect) showed comparatively lower precision (**0.65**) but maintained a good recall (**0.81**), indicating effective detection of borderline cases despite inherent ambiguity.

The confusion matrix further validates the model's effectiveness, with strong diagonal values: **307 Normal**, **48 Suspect**, and **32 Pathological instances correctly classified**. Misclassifications are minimal and mostly occur between adjacent classes (Normal–Suspect and Suspect–Pathological), which is expected due to overlapping clinical characteristics.

The ROC-AUC scores demonstrate excellent class separability, with values of **0.98 for Normal**, **0.95 for Suspect**, and **1.00 for Pathological**. The ROC curves are closely aligned to the top-left corner, indicating a high true positive rate and low false positive rate. Additionally, Precision-Recall curves show stable performance for Normal and Pathological classes across varying thresholds, while Suspect class exhibits a gradual precision decline at higher recall levels.

Feature importance analysis highlights that abnormal short-term variability**,** percentage of time with abnormal long-term variability, and accelerations are the most influential features contributing to model predictions. Histogram-based features and mean heart rate values also play a significant role, while lower-ranked features such as histogram tendency have minimal impact.

Comparison with Other Studies: A comparative analysis with existing studies indicates that the proposed model achieves competitive performance with an overall accuracy of **91%**, aligning closely with advanced machine learning and ensemble-based approaches reported in recent literature. While some deep learning models achieve marginally higher accuracy, they often require significantly higher computational resources and complex architectures. In contrast, the proposed approach offers a balanced trade-off between performance, efficiency, and interpretability, making it more suitable for practical clinical deployment.

B. Discussion

Clinical Implications:
The proposed system has significant clinical relevance as it supports early and accurate detection of fetal health conditions using CTG data, thereby assisting clinicians in timely decision-making. The high reliability in identifying pathological cases ensures that critical conditions can be flagged early, enabling prompt medical intervention and reducing the risk of adverse outcomes.

```
Classification Report:
              precision    recall  f1-score   support

           0       0.97      0.92      0.95       332
           1       0.65      0.81      0.72        59
           2       0.89      0.91      0.90        35

    accuracy                           0.91       426
   macro avg       0.84      0.88      0.86       426
weighted avg       0.92      0.91      0.91       426
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Fig 2. Classification Report

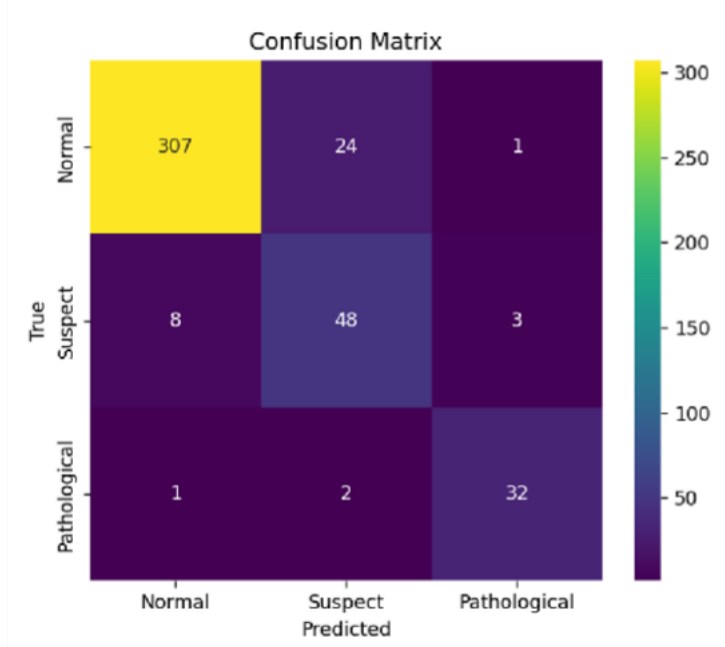

Fig 3. Confusion Matrix

Furthermore, the integration of explainable AI through SHAP enhances transparency, allowing healthcare professionals to understand the reasoning behind predictions and build trust in the system. This makes the model a valuable tool for clinical decision support, especially in resource-constrained settings where expert interpretation may not always be available.

The results indicate that the proposed system performs reliably across all fetal health categories, with particularly strong performance in identifying Normal and Pathological cases. The high accuracy and F1-score confirm that the model effectively captures complex patterns in cardiotocography data, making it suitable for clinical decision support.

The excellent performance of the Pathological class is especially significant, as accurate detection of high-risk fetal conditions is critical in medical applications. The ROC-AUC score of **1.00** for this class demonstrates near-perfect discrimination capability, ensuring that critical cases are identified with high confidence. This strengthens the system's potential for real-world deployment in assisting clinicians.

The comparatively lower precision for the Suspect class reflects the inherent complexity and ambiguity of borderline cases. These cases often share characteristics with both Normal and Pathological conditions, making them more difficult to classify. However, the model maintains a high recall for this class, ensuring that most suspect cases are still detected, which is clinically preferable to avoid missing potential risks.

The confusion matrix analysis shows that most misclassifications occur between neighboring classes, which is acceptable in medical classification problems where boundaries are not strictly defined. This behavior indicates that the model is learning meaningful relationships rather than making random errors.

The feature importance results align well with clinical knowledge, as variability-related features and heart rate patterns are known to be critical indicators of fetal health. The integration of explainability techniques further enhances trust in the model by providing transparent insights into decision-making.

Overall, the system demonstrates a strong balance between accuracy, interpretability, and robustness. While performance is highly satisfactory, further improvements can be achieved by incorporating larger and more diverse datasets, enhancing feature engineering, and optimizing model parameters to improve classification of borderline cases.

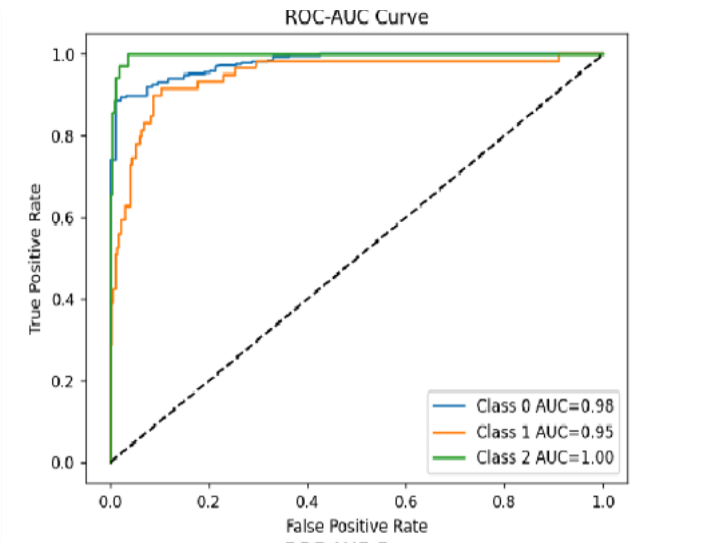

Fig 4. ROC-AUC Curve

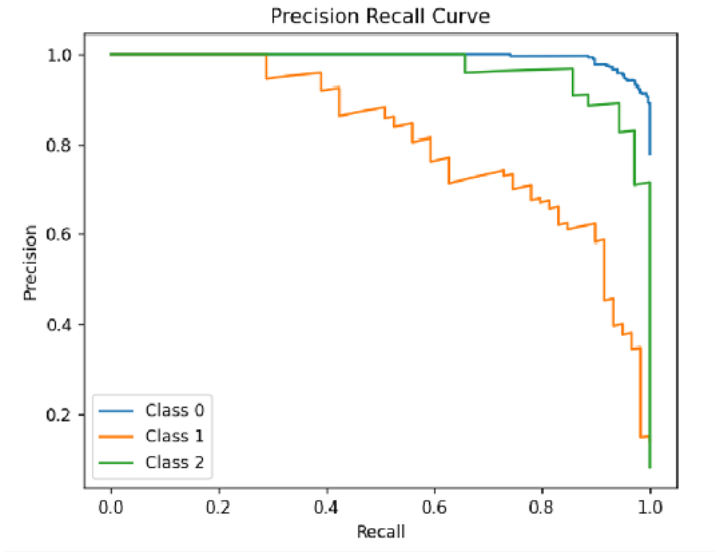

Fig 5. Precision – Recall Curve

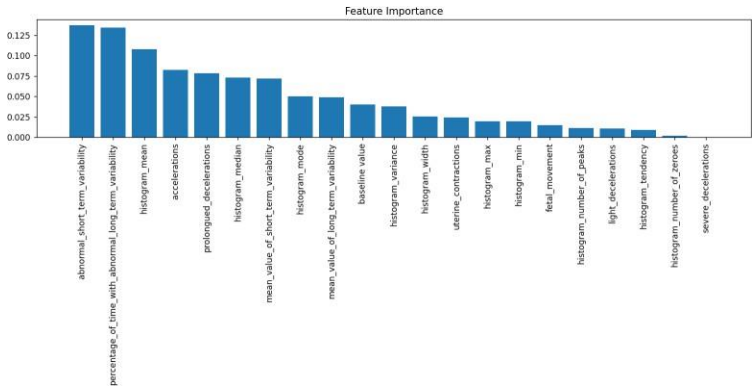

Fig 6. Feature Importance

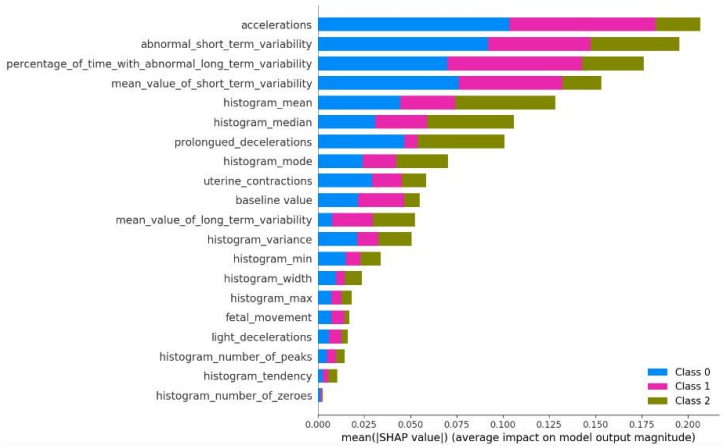

Fig 7. Global SHAP Visualizations

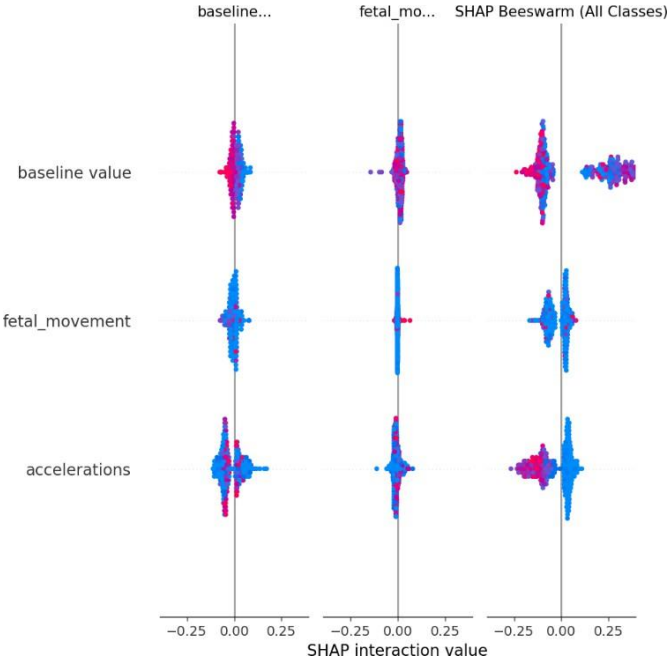

Fig 8. SHAP Beeswarm Analysis

Fig 9. Dashboard

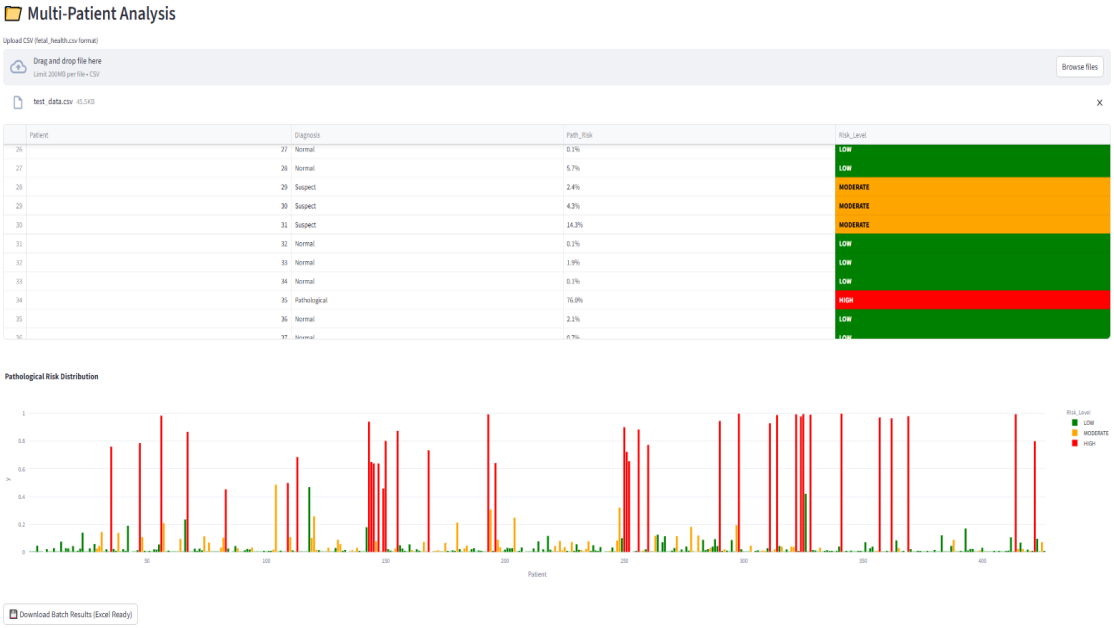

Fig 10. Multi-Patient Analysis & Pathological Risk Distribution

A	B	C	D	E
16	Normal	2.70%	LOW	
17	Normal	2.40%	LOW	
18	Normal	0.20%	LOW	
19	Normal	4.20%	LOW	
20	Normal	0.10%	LOW	
21	Normal	0.80%	LOW	
22	Normal	2.30%	LOW	
23	Normal	13.90%	LOW	
24	Normal	0.10%	LOW	
25	Normal	0.10%	LOW	
26	Normal	2.50%	LOW	
27	Normal	0.10%	LOW	
28	Normal	5.70%	LOW	
29	Suspect	2.40%	MODERATE	
30	Suspect	4.30%	MODERATE	
31	Suspect	14.30%	MODERATE	
32	Normal	0.10%	LOW	
33	Normal	1.90%	LOW	
34	Normal	0.10%	LOW	
35	Pathologic	76.00%	HIGH	
36	Normal	2.10%	LOW	
37	Normal	0.70%	LOW	
38	Suspect	13.70%	MODERATE	
39	Normal	0.10%	LOW	
40	Normal	2.00%	LOW	
41	Normal	0.70%	LOW	
42	Normal	18.90%	LOW	
43	Normal	0.00%	LOW	
44	Normal	0.10%	LOW	
45	Normal	0.10%	LOW	

Fig 11. Batch Results

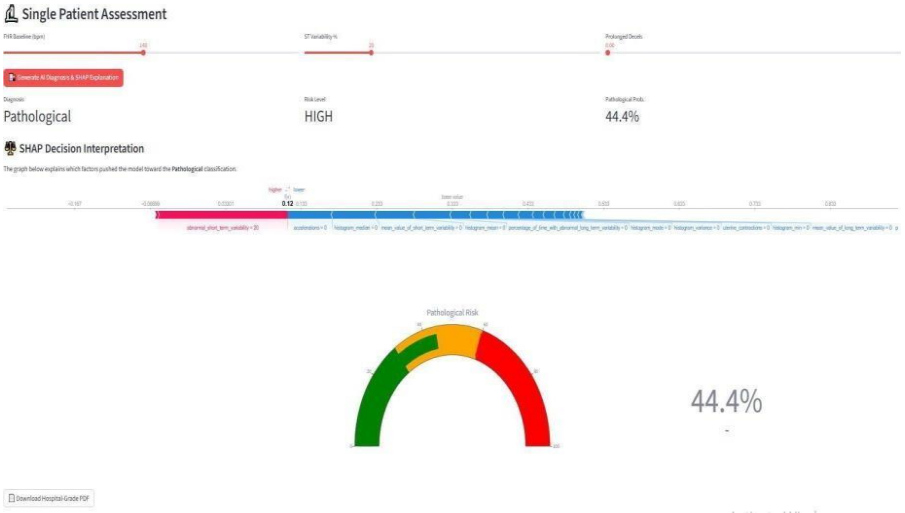

Fig 12. Single Patient Assessment & SHAP Decision Interpretation

Fetal Health Assessment Report

Cardiotocography (CTG) Analysis

PATIENT INFORMATION

Patient ID: 001 | **Date:** 19/03/2026 15:19

CTG PARAMETERS

Parameter	Measured	Normal Range
Baseline FHR (bpm)	140.00	110-160
Abnormal STV (%)	20.00	<50%
Prolonged Decels	0.00	0-0.1
Accelerations	0.00	>0.005
Light Decels	0.00	<0.01

AI FETAL HEALTH CLASSIFICATION

Primary Diagnosis: PATHOLOGICAL

Probability Distribution:
- Normal: **40.2%**
- Suspect: **15.4%**
- Pathological: **44.4%**

CLINICAL INTERPRETATION

PATHOLOGICAL FETAL STATUS

CRITICAL CTG abnormalities confirmed. Pathological risk: 44.4%

URGENT MANAGEMENT:
1. URGENT OBSTETRIC REVIEW
2. Stop oxytocin infusion
3. Left lateral position
4. 500ml IV fluids stat
5. Prepare for delivery if no improvement

KEY DECISION FACTORS

Rank	Feature	Model Weight
8	abnormal_short_term_variability	0.137
10	percentage_of_time_with_abnormal_long_term_variability	0.134
18	histogram_mean	0.108
2	accelerations	0.083
7	prolongued_decelerations	0.079

CTG FEATURE CLINICAL MAPPING

Feature	Clinical Meaning	High SHAP → Risk
abnormal_short_term_variability	Reduced FHR variability	Pathological (distress)
prolongued_decelerations	Late/variable decels	Suspect (hypoxia)
accelerations	Reactive FHR response	Normal (healthy)
baseline value	Mean FHR (bpm)	Extreme → Pathological

DISCLAIMER: AI supports clinical decision making per NICE/IPOC guidelines. Final decisions by attending clinician.

Fig 13. Assessment Report

Limitations: Despite achieving strong performance, the proposed model has certain limitations. The system is trained on a single CTG dataset, which may limit its generalizability across diverse populations and real-world clinical settings. The comparatively lower precision in the Suspect class indicates challenges in accurately classifying borderline cases. Additionally, the model relies solely on CTG features and does not incorporate other clinical parameters such as maternal history or ultrasound data, which could further enhance prediction accuracy.

Comparison with Other Prediction Methods:

Compared to traditional machine learning and deep learning approaches, the proposed method provides a balanced trade-off between accuracy, computational efficiency, and interpretability. While advanced deep learning models such as CNNs and transformers may achieve slightly higher accuracy, they require large datasets and high computational resources. In contrast, the Random Forest–based approach used in this study delivers competitive performance with lower complexity and, when combined with explainability techniques, offers greater transparency, making it more practical for real-world clinical decision support systems.

V. CONCLUSION

The proposed fetal health monitoring system successfully demonstrates the integration of machine learning and explainable artificial intelligence to provide an efficient and reliable decision support tool. By leveraging cardiotocography data, the system is capable of accurately classifying fetal conditions and presenting results through an intuitive interface. The implementation of a complete workflow—including data preprocessing, model training, prediction, visualization, and automated report generation—ensures that the system is not only technically sound but also practically usable in clinical scenarios.

In addition, the incorporation of explainability through SHAP enhances the transparency of the model, allowing users to understand the factors influencing each prediction. This improves trust and supports informed clinical decision-making. The system's ability to handle both individual and batch inputs further increases its applicability in real-world healthcare environments. With future improvements such as integration of additional clinical parameters and large-scale validation, the proposed solution has the potential to contribute significantly to intelligent and accessible fetal health monitoring systems.

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